AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM271
Machine Learning and Deep Learning

Final Examination
Spring Semester 2024/2025

Date: 11/June/ 2025

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 3 Hours

Instructions:

- Total Marks: **50**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **NOT** allowed.

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PART – ONE Total Marks 10

Question No. 1: Select the correct choice for each of the following statements.

1. What is the main difference between RNNs and traditional feedforward neural networks?

A) RNNs have loops that allow information to persist

B) RNNs are faster than feedforward networks

C) Feedforward networks process sequential data better

D) RNNs cannot process text data

2. What is the role of the decoder in an autoencoder?

A) It compresses input data

B) It reconstructs the original input from the latent space

C) It removes noise from the input

D) It extracts important features

3. Which component of a GAN is responsible for distinguishing real data from fake data?

A) Generator

B) Encoder

C) Discriminator

D) Decoder

4. Which of the following is an example of collaborative filtering?

A) Suggesting books based on genre and author

B) Recommending movies based on user ratings and behavior

C) Identifying similar users based on demographic data

D) Assigning categories to products using a decision tree

5. Which of the following is an example of NLP?

A) Image classification

B) Speech recognition

C) Sorting numbers in a list

D) Calculating mathematical equations

PART – TWO Total Marks 15

Question No. 2: What are activation functions? Explain any activation function with example **(4 Marks)**

Question No. 3: Explain why we use RNN? What can they do? Give at least three examples. **(6 Marks)**

Question No. 4: Explain the role of collaborative filtering in the context of recommended systems. **(5 Marks)**

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PART – THREE (Total Marks 25)

For the following questions, you are required to write the Python code.

Scenario: Predicting Air Quality Using Machine Learning

You are tasked with developing a regression model using a Deep Neural Network (DNN) to predict air quality index (AQI) based on environmental factors such as temperature, humidity, pollutant levels, and wind speed. You are provided with a dataset containing historical air quality measurements. Your objective is to design and train a DNN model to make accurate AQI predictions. Provide detailed Python code for this task, including comments to explain each step.

Question No 5: Loading the Dataset **(3 Marks)**

Write the Python code to load a dataset named "air_quality.csv" into a pandas DataFrame named aqi_data.

Question No 6: Dataset Splitting **(5 Marks)**

Write the Python code to normalize the dataset and divide it into training and test sets, assuming 80% for training and 20% for testing.

Question No 7: RNN Architecture Design **(12 Marks)**

Create DNN architecture with at least 3 layers and justify your choice of layers and loss function. Train your model and display the summary of the model.

Question No 8: Model Training and Prediction **(5 marks)**

Write the Python code to train the model and make predictions on the test dataset.

End of questions

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